

VME PRECAST WORKSPACE

Quick Reference SOP

Integrated Assembly Management & BBS Sync System

Plugin v1.1.0 | May 2026

STEEL (BBS)

CONCRETE

HOOKS & EMBEDS

1. INTERFACE OVERVIEW

UI Element	Action / Purpose
Assembly Name dropdown	Select the active precast assembly (e.g. PW-B1-30)
STEEL (BBS) tab	View rebar schedule, MTO summary & weights
CONCRETE tab	Concrete mix, volume & pour data
HOOKS & EMBEDS tab	Lifting anchors, plates & sleeve schedule
Sync All Assemblies (Blue)	Push all data to linked Revit sheets automatically
Export PDF (Red)	Export current assembly BBS as PDF
Batch Export All (Orange)	Export ALL assemblies as PDFs in one click
Status bar (bottom)	Shows active Revit Sheet (e.g. A-102)
? Help icon	Opens in-plugin documentation panel

■ Pro Tip

Type the first few letters of an assembly name in the dropdown to jump to it instantly — no need to scroll.

■ Pro Tip

The Status bar always shows which Revit Sheet is linked. If it's blank, the assembly has no sheet yet.

■ Pro Tip

Use the ? Help icon (top-right of the plugin) to access context-sensitive help for any active tab.

2. DAILY WORKFLOW

Step	Action
1 Select Assembly	Choose from the Assembly Name dropdown — all tabs refresh instantly.
2 Review Steel tab	Check Summary MTO (bar dia / total length / weight) and the full BBS table below.
3 Review other tabs	Switch to CONCRETE and HOOKS & EMBEDS tabs to verify data.
4 Sync	Click Sync All Assemblies to update all Revit sheets. Wait for confirmation.
5 Export	Click Export PDF for the current assembly, or Batch Export All for the full set.

■ Always Sync Before You Export

Running Export PDF without syncing first may produce outdated BBS data on your sheets. Always click Sync All Assemblies and wait for the confirmation message before exporting.

3. REBAR BENDING SCHEDULE — FIELD REFERENCE

Field	Meaning
Mark	Unique bar mark ID
Type	Bar function (e.g. T16 DOWEL BAR, T8 Link)

Size	Nominal diameter (mm)
Shape	BS 8666 code — 00 = straight, 26 = bent
A–E	Individual bending dimension legs (mm)
Bar Len	Length of one bar (mm)
Total Len	Bar Len × Qty
Qty	Number of bars
Weight	Total mass for this bar mark (kg) — shown in blue

■ **Shape Code 00**

Straight bar — no bends. All dimension fields B–E will show 0 mm.

■ **Shape Code 26**

Bent bar with multiple legs. Check all A–E dimensions against the structural drawing.

■ **Weight Column**

Highlighted in blue. Uses steel density 7850 kg/m³. Flag any value that looks unusually high or low.

4. BUTTON QUICK GUIDE

Sync All Assemblies	Updates every linked Revit sheet. Allow up to 5 min on large models. Do not close Revit.
Export PDF	Exports current assembly only. Choose save location in the dialog.
Batch Export All	Exports every assembly. Pick an output folder. Do not interact with Revit during export.

■ **Batch Export Timing**

Schedule Batch Export at end of day. Large projects (50+ assemblies) can take 10–15 min.

■ **PDF File Naming**

PDFs are saved as AssemblyName_BBS_Date.pdf by default. Set a project output folder once and reuse it.

5. TROUBLESHOOTING

Issue	Fix
Plugin tab missing in Revit	Reinstall as Administrator; verify Revit version compatibility
Assembly dropdown empty	Create assemblies in Revit model first, then reload plugin
BBS shows zero values	Check parameter mapping in plugin Settings
Sync fails	Ensure model is workshared and all assembly sheets exist
Batch Export stops mid-way	Check disk space & write permissions on output folder
Wrong weight values	Reset steel density to 7850 kg/m³ in Settings

6. KEY REMINDERS & SAFETY HINTS

■ **Lifting Anchors**

Always verify embed ratings against the element self-weight BEFORE casting. Under-rated anchors are a critical site safety risk.

■ **Sync First, Always**

Never submit a PDF that hasn't been synced in the same session. Data in Revit may have changed.

■ **Archive Exports**

Save dated exports to your project DMS (BIM 360, SharePoint). Do not rely on local desktop copies.

■ **Cross-check MTO**

Compare the Summary MTO weights against the structural engineer's take-off before ordering steel.

■ **Plugin Updates**

Check for VME Precast Workspace updates regularly. New versions may fix weight calculation or sync bugs.

■ **Assembly Naming**

Follow the PW-B1-XX convention consistently. Inconsistent names break the Batch Export sort order.